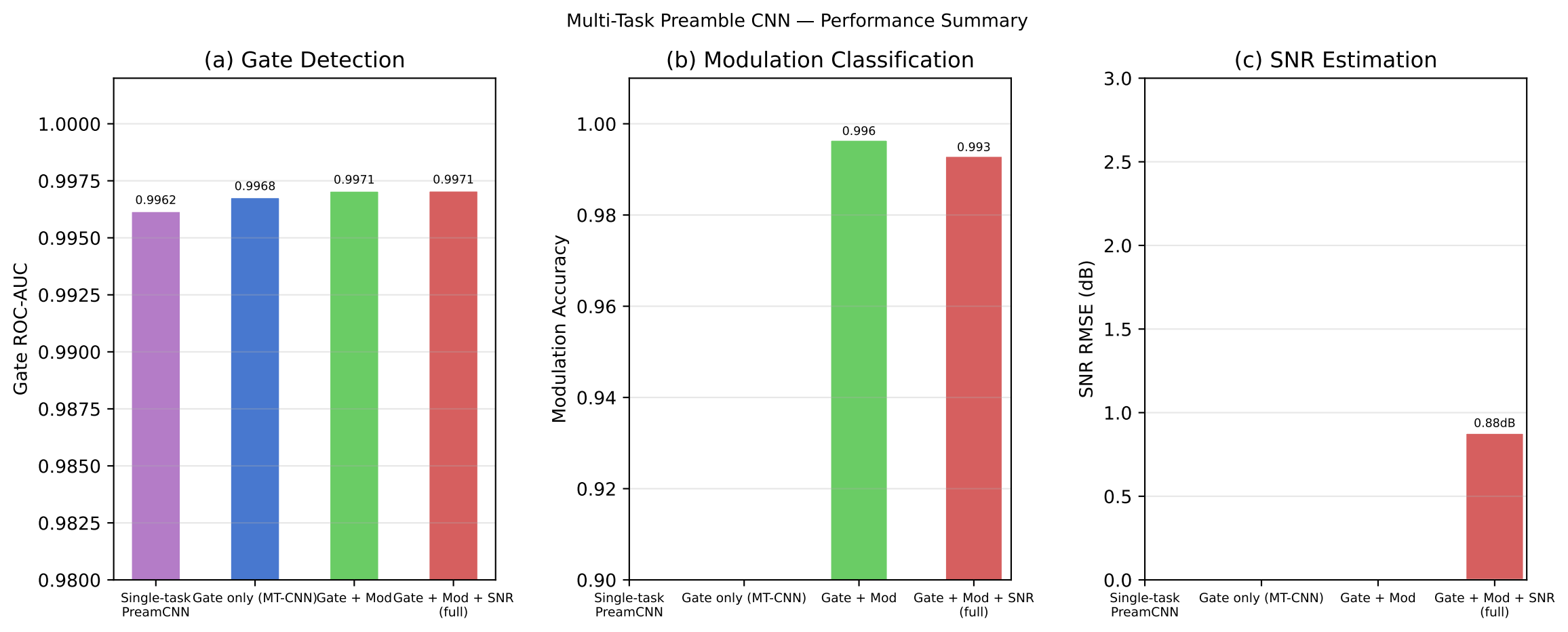
Multi-Task Preamble CNN — Performance Summary
(a) Gate Detection
Gate ROC-AUC
1.0000
0.9975
0.9950
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800
0.9962
0.9968
0.9971
0.9971
Single-task PreamCNN
Gate only (MT-CNN)
Gate + Mod
Gate + Mod + SNR (full)
(b) Modulation Classification
Modulation Accuracy
1.00
0.98
0.96
0.94
0.92
0.90
0.996
0.993
Single-task PreamCNN
Gate only (MT-CNN)
Gate + Mod
Gate + Mod + SNR (full)
(c) SNR Estimation
SNR RMSE (dB)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.88dB
Single-task PreamCNN
Gate only (MT-CNN)
Gate + Mod
Gate + Mod + SNR (full)